

Notes for the Lecture

1 Introduction

So far, we have assumed that players' payoffs are common knowledge. To paraphrase Harsanyi (1973) It might be more realistic to assume that players know their own payoff function, their knowledge of the other players' payoff function is somewhat inexact.¹

Harsanyi (1973) assumed that a player's payoff function is uncorrelated to its opponents' payoff functions. They are independent types. To understand his result, and the role of this assumption, let us start with a parametric example.

2 A Simple Example

This example follows Carlsson and van Damme (1993). Two players play a perturbed version of the normal-form game represented in Figure 1 below, where $x \in \mathbf{R}$.

	α_2	β_2
α_1	x, x	$x, 0$
β_1	$0, x$	$4, 4$

Figure 1: Game $g(x)$.

Given $X_0, X_1, X_2 \sim \mathcal{N}(0, 1)$, independently drawn, and $\delta, \varepsilon \in \mathbf{R}_+$, $x^* \in (0, 4)$, let

$$Y_i = x^* + \delta X_0 + \varepsilon X_i, \quad i = 1, 2.$$

¹Of course, to the extent that this is modeled here as a game of incomplete information with moves by Nature, the assumption of common knowledge becomes even more exorbitant—the exact details of the players' “ignorance” (the beliefs they assign to their opponents' possible payoff functions) is what is now assumed to be common knowledge.

Each player i is privately informed of y_i , the realization of Y_i . Given a strategy profile $s = (s_1, s_2)$, where s_i is the probability assigned to α_i , player i 's payoff is

$$g_i(y_i, s) = s_i y_i + 4(1 - s_1)(1 - s_2).$$

That is, values are private. If $\delta = 0$, types (in the sense of Harsanyi) are independent. If $\delta > 0$, values are positively correlated. We are particularly interested in what happens as $\varepsilon \downarrow 0$.

In this context, a pure strategy for player i is a measurable map

$$\begin{aligned} s_i: \mathbf{R} &\longrightarrow [0, 1] \\ y_i &\longmapsto s_i(y_i), \end{aligned}$$

specifying i 's probability of playing strategy α_i given signal y_i . Plainly, in any Bayes Nash equilibrium, $s_i(y_i) = 1$ if $y_i > 4$, and $s_i(y_i) = 0$ if $y_i < 0$. We look for symmetric cut-off equilibria, in which, for some $y \in [0, 4]$,

$$s_i(y_i) = \begin{cases} 0 & \text{if } y_i < y, \\ 1 & \text{if } y_i > y. \end{cases}$$

Because player i 's expected payoff is continuous in y_i , fixing $s_{-i}(\cdot)$, it must be that i is indifferent between both strategies at $y_i = y$. That is, it must hold that

$$4\mathbf{P}(Y_j < y \mid Y_i = y) = y.$$

Now,

$$Y_j \mid Y_i = y \sim \mathcal{N}(\mu, \sigma^2),$$

with

$$\mu := \frac{\delta^2 y + \varepsilon^2 x^*}{\delta^2 + \varepsilon^2}, \quad \sigma^2 := \frac{2\varepsilon^2 \delta^2 + \varepsilon^4}{\delta^2 + \varepsilon^2}.$$

That is, it must hold that

$$\Phi\left(\frac{y - \mu}{\sigma}\right) = \frac{y}{4},$$

where Φ is the cdf of the standard normal distribution. Equivalently,

$$\Phi(\eta(y - x^*)) = \frac{y}{4}, \tag{1}$$

where

$$\eta := \left(\frac{\varepsilon^2}{(2\delta^2 + \varepsilon^2)(\delta^2 + \varepsilon^2)} \right)^{\frac{1}{2}}.$$

One can show that (1) admits exactly one or three solutions, depending on $\eta \leq \eta(x^*)$.

Carlsson and van Damme do not prove this. Here is a proof. Plainly, at least one solution exists. Define $z := \eta(y - x^*)$. We are solving $F_\eta(z) = x^*$, where $F_\eta(z) := 4\Phi(z) - \frac{z}{\eta}$. Note that $F'_\eta(z) = 4\phi(z) - \frac{1}{\eta}$, where $\phi(z) := \Phi'(z)$. Hence, if $\eta < \frac{\sqrt{2\pi}}{4}$, F_η admits no critical point, and there can be at most one solution. Assume $\eta > \frac{\sqrt{2\pi}}{4}$, so $F'_\eta(z) = 0$ for exactly two values, $-z_0 < 0 < z_0$ (with $z_0 > 0$). Since $F_\eta(z) \rightarrow -\infty$ as $z \rightarrow \infty$ and $F_\eta(z) \rightarrow \infty$ as $z \rightarrow -\infty$, the function $F_\eta(z)$ has a local minimum at $-z_0$ and a local maximum at z_0 . Eqn. (1) admits three solutions if, and only if, $F_\eta(-z_0) < x^* < F_\eta(z_0)$. Since $\frac{1}{\eta} = 4\phi(z_0)$, $F_\eta(z_0) = 4(\Phi(z_0) - z_0\phi(z_0)) =: K(z_0)$. Similarly, $F_\eta(-z_0) = 4 - K(z_0)$. Hence, there are three solutions if, and only if, $4 - K(z_0) < x^* < K(z_0)$. Note that $K'(z) = 4z^2\phi(z) > 0$, $K(0) = 2$ and $\lim_{z \rightarrow \infty} K(z) = 4$. Wlog, given symmetry, assume $2 \leq x^* < 4$, so $4 - K(z_0) < x^*$. Since K is strictly increasing, there exists exactly one value $z(x^*)$ such that $K(z(x^*)) = x^*$. Recall that z_0 solves $4\phi(z_0) = \frac{1}{\eta}$. Hence, $z_0 > z(x^*)$ if, and only if, $\eta > \eta(x^*)$, where $\eta(x^*)$ is the unique solution to $\eta(x^*) = \frac{1}{4\phi(z(x^*))}$. Hence, there exists one, and only one, solution to (1) if, and only if, $\eta < \eta(x^*)$.

For $\eta \geq \eta(x^*)$, let $y^1(\eta) < y^2(\eta) < y^3(\eta)$ denote the solutions. Clearly,

$$\lim_{\eta \rightarrow \infty} y^1(\eta) = 0, \quad \lim_{\eta \rightarrow \infty} y^2(\eta) = x^*, \quad \lim_{\eta \rightarrow \infty} y^3(\eta) = 4.$$

Suppose first that $\delta, \varepsilon \rightarrow 0$, but also $\delta/\varepsilon \rightarrow 0$, so δ is vanishing relative to ε . Then $\eta \rightarrow \infty$. Also, since $Y_i \sim \mathcal{N}(x^*, \delta^2 + \varepsilon^2)$, the signals become arbitrarily precise. Hence, for the choice $y = y^1$, since $y^1 < x^*$, each player chooses α_i with probability that tends to 1 as $\eta \rightarrow \infty$. Similarly, if $y = y^3$, the outcome distribution tends to both players choosing β_i with probability 1. Finally, for $y = y^2$, player i chooses β_i with probability

$$\Phi\left(\frac{y^2 - x^*}{\sqrt{\delta^2 + \varepsilon^2}}\right) = \Phi\left(\eta(y^2 - x^*)\sqrt{\frac{2\delta^2 + \varepsilon^2}{\varepsilon^2}}\right) \rightarrow \frac{x^*}{4},$$

which is the probability with which β_i is played in unique mixed-strategy equilibrium of the game $g(x^*)$. This is Harsanyi's celebrated result, described below in more detail: every Nash equilibrium of the game under complete

information is outcome-equivalent to a pure-strategy equilibrium of the game with independently perturbed payoffs, as the perturbation vanishes, under conditions stated below.

If instead $\eta \rightarrow 0$, there is a unique cut-off equilibrium, with $y(\eta) \rightarrow 2$. Hence, in the limit, player i plays α_i if, and only if, $y_i > 2$. This is the *risk-dominant* strategy, namely, the one that is optimal against a strategy of the opponent that puts equal weight on both strategies.

3 Independent Types

Before proving Harsanyi's celebrated result more generally, it is necessary to introduce an important property of some Nash equilibria. Given some normal-form game (S, u) , it is often useful to define the *extension* of the game by enlarging the domain of a player's payoff function $u_i(s_i, \cdot)$ from Σ_{-i} to all $\times_{j \neq i} \mathbf{R}^{|S_j|}$ the obvious way, namely, for all $\sigma_{-i} \in \times_{j \neq i} \mathbf{R}^{|S_j|}$,

$$u_i(s_i, \sigma_{-i}) = \sum_{s_{-i}} \sigma_{-i}(s_{-i}) u_i(s_i, s_{-i}).$$

We set $m = \sum_{i=1}^n |S_i|$, and use k as a generic superscript for a given strategy. Hence, $\sigma \in \mathbf{R}_+^m$ is a Nash equilibrium if, and only if, it solves the system

$$\begin{cases} \sum_k \sigma_i(s_i^k) - 1 = 0, \text{ all } i, \\ \sigma_i(s_i^k) \left(u_i(s_i^k, \sigma_{-i}) - \max_{k'} u_i(s_i^{k'}, \sigma_{-i}) \right) = 0, \text{ all } i, s_i^k \in S_i. \end{cases}$$

This is a system of $m + n$ equations in m unknowns, but then again since the maximum must be achieved by some k' , there are n equations that are trivially satisfied, and so this is really a system of m equations in m unknowns. What is inconvenient is the maximum that appears in this system. Hence, it is more convenient to work with, fixing each player's first strategy as a reference,

$$\begin{cases} \sum_k \sigma_i(s_i^k) - 1 = 0, \text{ all } i, \\ \sigma_i(s_i^k) \left(u_i(s_i^k, \sigma_{-i}) - u_i(s_i^1, \sigma_{-i}) \right) = 0, \text{ all } i, s_i^k \in S_i. \end{cases}$$

So, every Nash equilibrium σ for which $(s_1^1, \dots, s_n^1)$ is in the support of σ is a solution of this system. However, not every nonnegative solution to this system is a Nash equilibrium to this system, and not every Nash equilibrium solves this system. But the right-hand side involves a mapping (from $\mathbf{R}^m$ into $\mathbf{R}^m$) that is infinitely differentiable. Hence we introduce $f : \mathbf{R}^m \rightarrow \mathbf{R}^m$ as

$$\begin{cases} f_{i1}(\sigma) = \sum_k \sigma_i(s_i^k) - 1, & \text{all } i, \\ f_{ik}(\sigma) = \sigma_i(s_i^k) (u_i(s_i^k, \sigma_{-i}) - u_i(s_i^1, \sigma_{-i})), & \text{all } i, k > 1. \end{cases}$$

In what follows, as a convention, we take for granted that every Nash equilibrium σ considered is such that $(s_1^1, \dots, s_n^1)$ is in the support of σ .

Definition 1 *A Nash equilibrium $\sigma \in \Sigma$ is a regular equilibrium if the Jacobian $J(\sigma)$ is nonsingular.*

Recall that the Jacobian is the matrix of first-order partial derivatives, that is,

$$J(\sigma) = \left. \frac{\partial f(\sigma')}{\partial \sigma'} \right|_{\sigma'=\sigma}.$$

It is important to note that the choice of the pure strategy profile s used as a reference in the definition of f (here, $(s_1^1, \dots, s_n^1)$) is irrelevant, as long as it is in the support of σ . Regularity is a property independent of that choice.

An important property of regular equilibria is that they are *quasi-strict*: if σ is regular, then $\sigma^i(s_i^k) = 0 \Rightarrow u_i(s_i^k, \sigma_{-i}) - u_i(s_i^1, \sigma_{-i}) < 0$. If an action isn't played with positive probability, it is strictly suboptimal.

A fundamental result that is an immediate consequence from Sard's theorem is the following:

Theorem 1 *Generically, all Nash equilibria of normal-form games are regular.*

Here, “generically” refers to the payoff vector $u \in \mathbf{R}^m$ (wlog, we may normalize $\|u\| = 1$). Of course, many normal-form games of interest are not generic (extensive forms may lead to payoffs in the normal form with “ties”). And indeed, there are elegant examples showing that the result we prove below

does not extend to (the normal form that results from) some extensive forms, for generic payoffs assigned to (the terminal nodes of) those extensive forms.

Fix a regular Nash equilibrium $\bar{\sigma}$. By the implicit function theorem, and continuity of $u(\cdot)$, there exists an open $U \subset \mathbf{R}^m$ such that, for some $\varepsilon > 0$,

$$\begin{aligned} \bar{\sigma} &\in U, \\ \|\bar{\sigma} - \sigma\| &< \varepsilon, \text{ all } \sigma \in U, \\ \text{cl } U \cap f^{-1}(0) &= \{\bar{\sigma}\}, \\ \bar{\sigma}_i(s_i^k) > 0 &\Rightarrow \sigma_i(s_i^k) > 0, \text{ all } i, k, \\ u_i(s_i^k, \bar{\sigma}_{-i}) - u_i(s_i^1, \bar{\sigma}_{-i}) < 0 &\Rightarrow u_i(s_i^k, \sigma_{-i}) - u_i(s_i^1, \sigma_{-i}) < 0, \text{ all } i, k, \sigma \in \text{cl } U. \end{aligned}$$

Before proving the main result of this section, we need to state an elegant result from algebraic topology.²

Proposition 1 *Given a bounded open set $U \subset \mathbf{R}^m$, and continuous maps $f, h : \text{cl } U \Rightarrow \mathbf{R}^m$, suppose that (1) f is continuously differentiable on U , and (2) x_0 is the only zero of f in U , and that the Jacobian determinant of f at x_0 is nonzero. If, for every $t \in [0, 1]$, the function $th + (1 - t)f$ has no zero on $\text{bd } U$, then h has a zero in U .*

Given the normal-form game $G = (S, u)$, define the perturbed game of incomplete information $G^\eta = (S, u^\eta, \mu_1, \dots, \mu_n)$, where player i 's payoff is defined by $u_i(s) + \eta_i(s)$, where $\eta_i \in \mathbf{R}^{|S|}$ is drawn according to μ_i , independently for $i = 1, \dots, n$. Here, μ_i is a Borel probability measure on $\mathbf{R}^{|S|}$. We assume that μ_i is absolutely continuous with respect to Lebesgue measure on $\mathbf{R}^{|S|}$. Denote by δ_0 the Dirac measure at $\mathbf{0}$ in $\mathbf{R}^{|S|}$.

Harsanyi's result is the following.

Theorem 2 *Suppose $\bar{\sigma}$ is a regular Nash equilibrium of G . Consider any sequence $\mu^{n'}$ such that, for all i , $\mu_i^{n'} \rightarrow \delta_0 \in \mathbf{R}^{|S|}$. For all $\varepsilon > 0$, there exists N , for all $n' > N$, the game $G^\eta = (S, u^\eta, \mu_1^{n'}, \dots, \mu_n^{n'})$ has a strict (up to a $\mu^{n'}$ -measure zero set of values of η) Nash equilibrium inducing a distribution on S that is within ε of $\bar{\sigma}$.*

We note that every equilibrium of the perturbed game is essentially strict (as defined in the theorem), and although we won't show this, every limit point

²See McLennan (1989), for instance.

of a sequence of equilibria of the perturbed game is an equilibrium of the unperturbed game.

To see that the equilibrium must be essentially strict, note that, for any $s_i, s'_i \in S_i$, player i is indifferent between them if, and only if,

$$\sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i})(\eta_i(s'_i, s_{-i}) - \eta_i(s_i, s_{-i})) = \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i})(u_i(s_i, s_{-i}) - u_i(s'_i, s_{-i})),$$

but the probability under μ_i that for a fixed σ_{-i} , the left-hand side takes the fixed value on the right-hand side is zero.

This implies that, μ_i -a.s.,

$$b_i(\sigma, \eta_i) = \arg \max_{s_i} \sum_{s_{-i} \in S_{-i}} \sigma_{-i}(s_{-i})(u_i(s_i, s_{-i}) + \eta_i(s_i, s_{-i}))$$

is a singleton: the best-reply is pure and strict. Let $g_i(\sigma) = \int_{\mathbf{R}^{|S|}} b_i(\sigma, \eta_i) d\mu_i(\eta_i)$ denote the *expected* best reply. Note that $g_i(\cdot)$ is a continuous function, and that the Bayes Nash equilibria of G^n are precisely the fixed-points of the map $g = (g_1, \dots, g_n)$. Define similarly $g^{n'}$.

To prove the proposition, it suffices to show that $h^{n'}$ has a zero in U for all n' large enough, where $h^{n'} : \text{cl } U \Rightarrow \mathbf{R}^m$ is defined as

$$\begin{cases} h_{i1}^{n'}(\sigma) = \sum_k \sigma_i(s_i^k) - 1, & \text{all } i, \\ h_{ik}^{n'}(\sigma) = g_i^{n'}(s_i^k, \sigma_{-i}) - \sigma_i(s_i^k), & \text{all } i, k > 1, \end{cases}$$

because σ a zero of this system is obtained if $g^{n'}(\sigma_i) = \sigma_i$, all i , that is, σ is a fixed-point of $g^{n'}$.

Given Proposition 1, it suffices to show that:

Lemma 1 *There exists N , for all $n' > N$, and for all $t \in [0, 1]$, $th^{n'} + (1-t)f$ has no zero on $\text{bd } U$.*

Proof. (Govindan, Reny and Robson, 2003). Suppose not, that is, there exists $t^{n'} \in [0, 1]$, $t^{n'} \rightarrow \hat{t}$, $\sigma^{n'} \in \text{bd } U$, $\sigma^{n'} \rightarrow \hat{\sigma}$, such that $\sigma^{n'}$ is a zero of $t^{n'}h^{n'} + (1-t^{n'})f$.

1. Since u_i is continuous, $u_i(s_i^k, \hat{\sigma}_{-i}) - u_i(s_i^1, \hat{\sigma}_{-i}) < 0 \Rightarrow g_i^{n'}(s_i^k, \sigma_{-i}^{n'}) \rightarrow 0$

(by definition of a best-reply). Therefore, if $u_i(s_i^k, \hat{\sigma}_{-i}) - u_i(s_i^1, \hat{\sigma}_{-i}) < 0$,

$$\begin{aligned} 0 &= \lim_{n'} t^{n'} h_{ik}^{n'}(\sigma^{n'}) + (1 - t^{n'}) f_{ik}(\sigma^{n'}) \\ &= -\hat{t} \hat{\sigma}_i(s_i^k) + (1 - \hat{t}) f_{ik}(\hat{\sigma}) \\ &= -\hat{\sigma}_i(s_i^k) (\hat{t} + (1 - \hat{t}) (u_i(s_i^1, \hat{\sigma}_{-i}) - u_i(s_i^k, \hat{\sigma}_{-i}))) \end{aligned} \quad (2)$$

implies that $\hat{\sigma}_i(s_i^k) = 0$.

2. $\hat{\sigma} \in \Sigma$: To see this, note first that if $\hat{\sigma}_i(s_i^k) < 0$, then $\bar{\sigma}_i(s_i^k) = 0$ (since $\bar{\sigma}_i(s_i^k) > 0 \Rightarrow \sigma_i(s_i^k) > 0$ for all $\sigma \in \text{cl } U$). Since $\bar{\sigma}$ is quasi-strict, this implies $u_i(s_i^k, \bar{\sigma}_{-i}) - u_i(s_i^1, \bar{\sigma}_{-i}) < 0$ and so $u_i(s_i^k, \hat{\sigma}_{-i}) - u_i(s_i^1, \hat{\sigma}_{-i}) < 0$. By step 1, this implies $\hat{\sigma}_i(s_i^k) = 0$, a contradiction. So, $\hat{\sigma}_i(s_i^k) \geq 0$, all i, k .

To see that the probabilities add up to one, note that $h_{i1}^{n'} = f_{i1}$, and so

$$\begin{aligned} 0 &= t^{n'} h_{i1}^{n'} + (1 - t^{n'}) f_{i1}^{n'} \\ &= f_{i1}(\sigma^{n'}), \end{aligned}$$

and so $\sum_k \sigma_i^{n'}(s_i^k) = 1$, all n' , and hence $\sum_k \hat{\sigma}_i(s_i^k) = 1$.

3. It holds that

$$\sum_{s_i^{k'} \in B_i(\hat{\sigma})} f_{ik'}(\sigma^{n'}) > 0,$$

where $B_i(\sigma) := \arg \max_{s_i^k} u_i(s_i^k, \sigma_{-i})$. Because $\hat{\sigma}$ is not a zero of f (only $\bar{\sigma}$ is a zero of f on $\text{cl } U$), there exists i, k such that $f_{ik}(\hat{\sigma}) \neq 0$. Point 1 then implies that $u_i(s_i^1, \hat{\sigma}_{-i}) - u_i(s_i^k, \hat{\sigma}_{-i}) < 0$, and so $s_i^1 \notin B_i(\hat{\sigma})$. Also, for all $s_i^{k'} \in B_i(\hat{\sigma})$, $\hat{\sigma}_i(s_i^{k'}) > 0$, since $B_i(\hat{\sigma}) \subset B_i(\bar{\sigma})$, and $\bar{\sigma}$ is quasi-strict. Hence the conclusion holds for all large enough n' .

4. It holds that

$$\sum_{s_i^{k'} \in B_i(\hat{\sigma})} h_{ik'}^{n'}(\sigma^{n'}) > 0.$$

To see this, note that $\hat{\sigma}_i(s_i^1) > 0 \Rightarrow \sum_{s_i^{k'} \in B_i(\hat{\sigma})} \hat{\sigma}_i(s_i^{k'}) < 1$, on one hand, and on the other, since $s_i^{k'} \notin B_i(\hat{\sigma}) \Rightarrow g_i^{n'}(s_i^{k'}, \sigma_{-i}^{n'}) \rightarrow 0$, it must be that $\sum_{s_i^{k'} \in B_i(\hat{\sigma})} g_i^{n'}(s_i^{k'}, \sigma_{-i}^{n'}) \rightarrow 1$.

5. Steps 3 and 4 directly contradict that $\sigma^{n'}$ is a zero of $t^{n'} h^{n'} + (1 - t^{n'}) f$.

■

This is one of the most convincing interpretations of mixed strategies: randomizations are not actual lotteries implemented by the players, but they arise from small uncertainties regarding players' payoffs, in terms of the distribution over pure strategies that this uncertainty implies. But the players always have a strict incentive to play one or the other pure strategy. This is referred to as *purification*. (Note, however, that pure Nash equilibria of the unperturbed game are also obtained as limits: purification does not select mixed-strategy equilibria, but it explains how we may interpret those as pure equilibria of a “nearby” game.)

There are two alternative interpretations of mixed strategies that are worth mentioning here. First, when focusing on two-person zero-sum games, we may interpret mixing as an act of *deliberate* randomization whose purpose is to conceal.

Thus one important consideration for a player in such a game is to protect himself against having his intentions found out by his opponent. Playing several such strategies at random, so that only their probabilities are determined is a very effective way to achieve a degree of such protection: By this device the opponent cannot possibly find out what the player's strategy is going to be, since the player does not know it himself. — von Neumann and Morgenstern (1944)

It is difficult to subscribe to this interpretation in other classes of games, coordination games, for instance: why conceal?

By eliminating the payoff perturbations altogether, we might directly interpret a player's mixed strategy solely as an expression of the other players' uncertainty about that player's pure strategy choice.

In recent years, a different view of mixing has emerged. According to this view, players do not randomize; each player chooses some definite action. But other players need not know which one, and the mixture represents their uncertainty, their conjecture about his choice. — Aumann and Brandenburger (1995)

While it is an attractive interpretation, I do not view it as particularly convincing: just as it seems implausible that a player randomizes in a way that makes her opponent indifferent across pure strategies in the support of her best-reply (what would be the reason for this?), there seems to be no compelling reason for why a player's perception of her opponent's pure strategy would happen to be the one that makes the player indifferent herself. Of course, we might interpret it as arising from a perturbation of the game, which formally captures the uncertainty that players entertain, but that is precisely Harsanyi's argument.

4 Global Games

Following Carlsson and van Damme (1993), we consider 2×2 normal-form games, and denote by $G = \mathbf{R}^8$ the set of all such games, identified with the payoff bimatrix, and let $S_i = \{\alpha_i, \beta_i\}$. Write $u_i(\sigma)$ for the payoff of player i from assigning probability σ to action α_i . Let

$$g_i^\alpha = u_i(\alpha) - u_i(\beta_i, \alpha_{-i}), \quad g_i^\beta = u_i(\beta) - u_i(\alpha_i, \beta_{-i})$$

denote the loss from deviating from $\alpha = (\alpha_1, \alpha_2)$ and $\beta = (\beta_1, \beta_2)$. let

$$G^\gamma = \{g \in G \mid g_i^\gamma > 0, \quad i = 1, 2\}$$

denote the games for which $\gamma \in \{\alpha, \beta\}$ is a strict Nash equilibrium, and

$$D_i^\alpha = \{g \in G^\alpha \mid g_i^\beta < 0, \quad i = 1, 2\}, \quad D_i^\beta = \{g \in G^\beta \mid g_i^\alpha < 0, \quad i = 1, 2\}$$

the games in which it α or β are the strictly dominant actions for i . Write

$$D^\gamma = D_1^\gamma \cup D_2^\gamma$$

for the games for which γ is uniquely selected by iterated deletion of strictly dominated strategies. For $g \in G^\alpha \cap G^\beta$, α *risk-dominates* β if

$$g_1^\alpha g_2^\alpha > g_1^\beta g_2^\beta,$$

that is, if the product of deviation losses is larger at α than at β . It is readily seen that this is equivalent to

$$\bar{\sigma}_1 + \bar{\sigma}_2 < 1,$$

where

$$\bar{\sigma}_{-i} = \frac{g_i^\beta}{g_i^\alpha + g_i^\beta}$$

is the probability with which $-i$ should play α_{-i} to make player i indifferent across actions. In a symmetric game, the risk-dominant strategy is the one one would prefer if her opponent was putting equal weight on both strategies.

Let R^α denote the set of games for which α is either the unique strict equilibrium or risk-dominates β :

$$R^\alpha = \{g \in G^\alpha \mid g_1^\beta, g_2^\beta > 0 \Rightarrow \bar{\sigma}_1 + \bar{\sigma}_2 < 1\},$$

and similarly for R^β .

The Bayesian game Γ^ε unfolds as follows.

1. Nature draws the game $x \in \Theta$, where $\Theta \subset \mathbf{R}^m$, for some m , according to some distribution with strictly positive, continuously differentiable and bounded density h .
2. Player i privately observes

$$X_i^\varepsilon = X + \varepsilon E_i,$$

for some $\varepsilon > 0$, where (E_1, E_2) are r.v. independent of X with joint density ϕ , s.t. $\text{supp} E_i \subset \mathcal{B}(0, 1)$, the unit ball in $\mathbf{R}^8$.

3. Players choose actions.

Payoffs are determined by $g(x)$, where $g : \Theta \rightarrow \mathbf{R}^8$, with $\frac{\partial g^k}{\partial x^l}$ continuous and bounded, for all $k = 1, \dots, 8$, $l = 1, \dots, m$.

A strategy in this Bayesian game is a map $\sigma_i : \mathbf{R}^m \rightarrow \Delta(\{\alpha, \beta\})$, for each possible type x_i .

The main result is the following.

Theorem 3 *If $x \in \mathcal{C}$, where $\mathcal{C} \subset \Theta$ is a curve s.t. $g(\mathcal{C}) \subset R^\gamma$, $g(\mathcal{C}) \cap D^\gamma \neq \emptyset$, then there exists $\bar{\varepsilon} > 0$, for all $\varepsilon < \bar{\varepsilon}$, γ is the unique action obtained by iterated elimination of strictly dominated strategies in Γ^ε when $X = x$.*